

Results Overview

Deep RL for Optimal Order Execution on Hyperliquid Order-Book Data

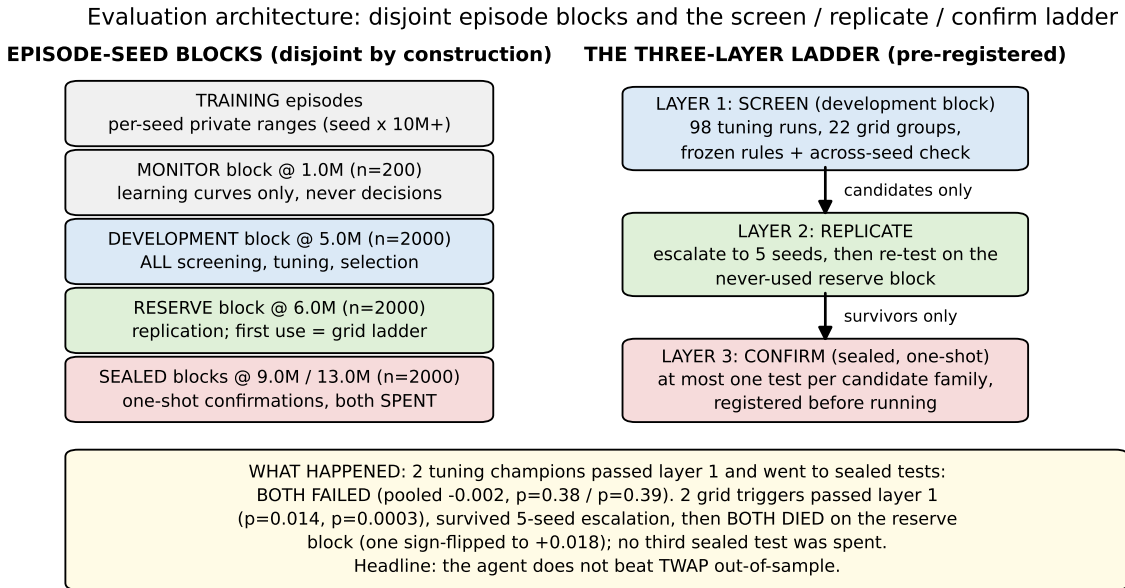
Fardeen Idrus

17 July 2026

One-paragraph summary. Across two market models, a full hyperparameter search, and a 4×4 size-by-deadline grid, no DRL agent beats TWAP out of sample. The strongest in-sample signals (up to $p = 0.0001$) were shown to be evaluation-set artifacts by one-shot sealed tests and a replication block. DQN fails for a diagnosed, systematic reason. The methodology that caught all of this is itself a central contribution.

1. Setup and validation

Evaluation architecture



- Disjoint episode blocks by construction: training ranges (private per seed), a monitor block (learning curves only), the development block (all screening and selection), a reserve block (replication), and sealed blocks (one-shot confirmations).
- Three layers: screen on development data, replicate survivors at five seeds on the reserve block, confirm once on sealed data.
- Rationale: roughly 200 configurations were screened on the same development episodes; under that selection pressure a single validation set is partially used up (winner's curse), so a replication layer is inserted, as in discovery and replication cohorts.

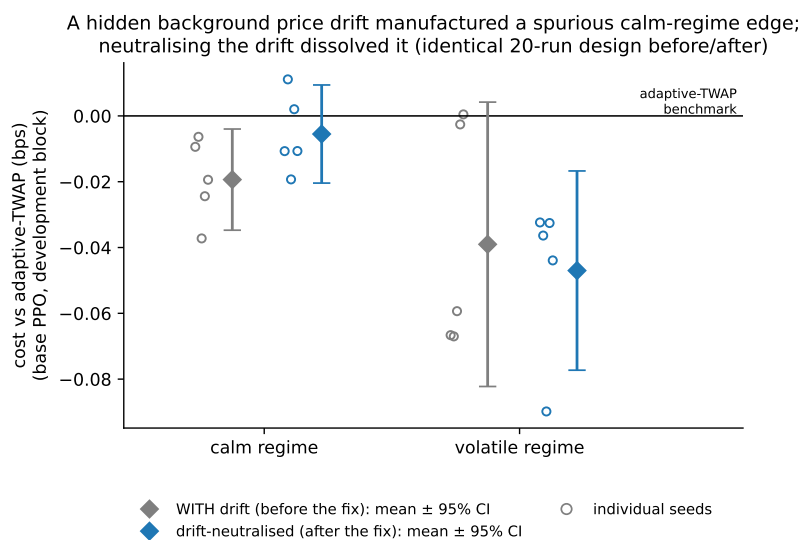
- Outcome so far: two champions reached sealed tests and both failed; two grid triggers died on the reserve block; no sealed test was wasted on them.
- The replication layer was added mid-project, dated and documented, after the first two sealed failures; its first use justified it immediately.

Environment validation gates (table)

regime	check	measured	pass band	verdict
calm	own impact is real and persists (2nd immediate dump / 1st, cost ratio)	2.41	≥ 1.25	PASS
calm	dump cost increases with order size (Spearman ρ)	1.00	> 0	PASS
calm	book refills after impact (probe cost +30s vs +1s, bps)	-0.46 vs 0.35	lower at +30s	PASS
calm	cost-vs-size growth matches the real book (sim vs real ratio)	2.45 vs 2.46	within band	PASS
calm	fixed-TWAP completes every episode	100%	$\geq 99\%$	PASS
calm	dumping costs more than scheduling (dump / TWAP, drift-free, bps)	0.40 / 0.20	dump $\geq$ TWAP	PASS
calm	no residual background drift (ticks/episode; t)	0.12 ($t=0.27$)	t n.s.	PASS
calm	no constant-pace policy beats TWAP (max $ t $ across paces)	1.42	none significant	PASS
volatile	own impact is real and persists (2nd immediate dump / 1st, cost ratio)	2.35	≥ 1.25	PASS
volatile	dump cost increases with order size (Spearman ρ)	1.00	> 0	PASS
volatile	book refills after impact (probe cost +30s vs +1s, bps)	-0.23 vs 0.14	lower at +30s	PASS
volatile	cost-vs-size growth matches the real book (sim vs real ratio)	2.44 vs 1.99	within band	PASS
volatile	fixed-TWAP completes every episode	100%	$\geq 99\%$	PASS
volatile	dumping costs more than scheduling (dump / TWAP, drift-free, bps)	0.24 / 0.05	dump $\geq$ TWAP	PASS
volatile	no residual background drift (ticks/episode; t)	0.07 ($t=0.06$)	t n.s.	PASS
volatile	no constant-pace policy beats TWAP (max $ t $ across paces)	1.77	none significant	PASS

- All gates pass in both regimes, after the drift correction.
- Self-impact is real and persistent: executing the 25 BTC order makes an identical immediate follow-on roughly $2.4\times$ more expensive; dump costs are monotone in size.
- The book demonstrably refills: a probe order is cheaper 30 seconds after the dump than 1 second after.
- Simulated cost-versus-size growth matches the real books (calm: 2.45 vs 2.46).
- Fairness gate: residual background drift statistically zero; no constant-pace policy is significantly cheaper than TWAP (max $|t| = 1.77$), so the benchmark race is fair by construction.

The drift confound, found and fixed



- December 2025 was a rising month; naive replay of its statistics made prices drift upward in almost every episode, mechanically rewarding fast buying.
- Caught by an adversarial pre-launch audit before any conclusion was drawn.
- Fix: drift-neutralised background moves with volatility, spread and impact preserved.
- The figure shows an identical 20-run design before and after (agents retrained under each background process): the calm-regime “edge” shrinks from -0.019 bps to an immaterial -0.006 bps, statistically indistinguishable from zero and an order of magnitude below the 0.05 bps materiality floor.
- A permanent fairness gate now guards this class of bug.

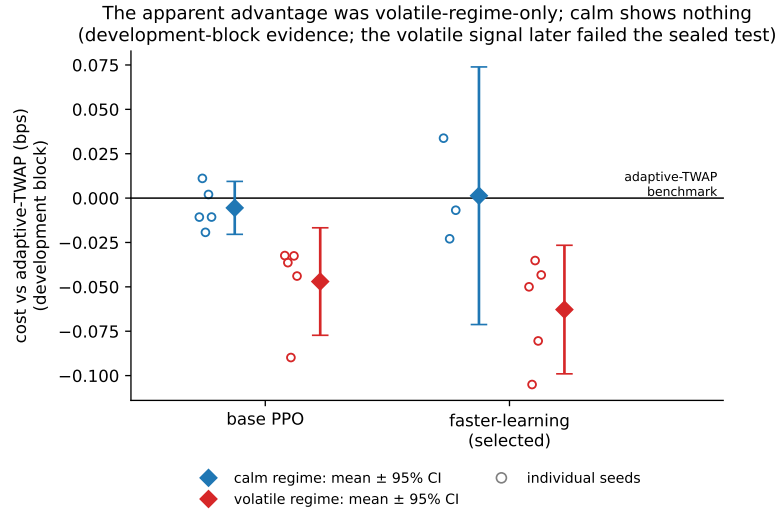
2. Can DRL beat TWAP? The headline

Primary campaign, all 20 runs (table)

algo	regime	seed	vs fixed-TWAP (bps)	p_{fix}	vs adaptive-TWAP (bps)	valid
dqn	calm	0	+0.1124	0.015	+0.1125	NO
dqn	calm	1	+0.0596	0.072	+0.0597	NO
dqn	calm	2	+0.0263	0.326	+0.0264	NO
dqn	calm	3	+0.1533	0.000	+0.1534	NO
dqn	calm	4	-0.0016	0.829	-0.0015	yes
dqn	volatile	0	+0.4176	0.001	+0.4169	NO
dqn	volatile	1	-0.0296	0.632	-0.0303	yes
dqn	volatile	2	+0.3017	0.003	+0.3011	NO
dqn	volatile	3	-0.0389	0.383	-0.0396	yes
dqn	volatile	4	-0.0197	0.893	-0.0203	NO
ppo	calm	0	-0.0108	0.958	-0.0107	yes
ppo	calm	1	+0.0110	0.576	+0.0111	yes
ppo	calm	2	-0.0194	0.324	-0.0193	yes
ppo	calm	3	+0.0020	0.597	+0.0020	yes
ppo	calm	4	-0.0108	0.632	-0.0107	yes
ppo	volatile	0	-0.0317	0.350	-0.0324	yes
ppo	volatile	1	-0.0357	0.442	-0.0364	yes
ppo	volatile	2	-0.0892	0.012	-0.0898	yes
ppo	volatile	3	-0.0319	0.417	-0.0326	yes
ppo	volatile	4	-0.0433	0.444	-0.0439	yes

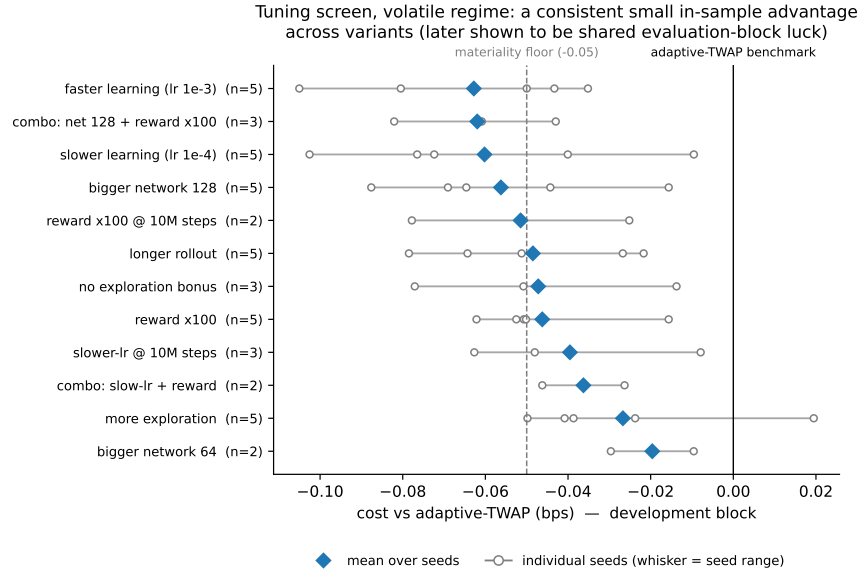
- PPO and DQN, calm and volatile, five seeds each, 25 BTC over five minutes, one decision per second.
- No cell meets the frozen edge definition. That definition was fixed before any run and has four parts: the agent must be cheaper than both versions of TWAP; each individual run must be statistically strong on its own (under one-in-a-hundred odds of chance); the direction must agree in at least four of the five runs; and the average saving must be at least 0.05 basis points.
- All ten PPO runs pass the behaviour audit; seven of ten DQN runs are invalid (collapse; Section 4).
- The interesting sub-threshold structure: volatile PPO is cheaper in five of five seeds.
- Why that is still not a headline edge: five agreeing directions alone happen by chance about one time in 32; the average saving (0.047) sits below the 0.05 floor; no single run is individually significant; and, decisively, the same signal later returned zero on both sealed tests (Section 2 below).

Regime split of the development signal



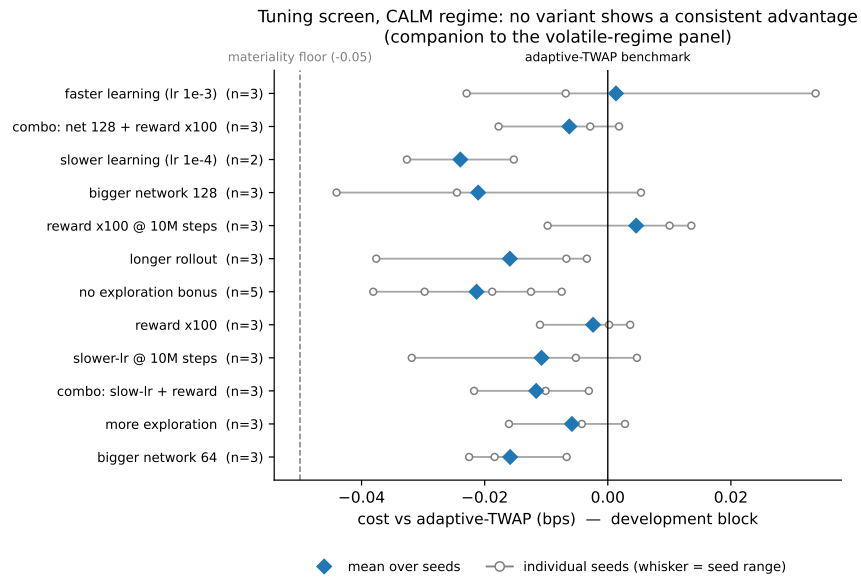
- After the drift fix, calm is a tie; the volatile PPO group pools -0.047 bps with across-seed $p = 0.006$.
- This volatile signal is the candidate edge that everything below chased.
- Research question 2 note: the signal is regime-specific from the very first campaign.

Tuning screen, volatile regime



- Twelve pre-registered variants, escalations to five seeds for the leaders.
- Seven of eight healthy variants reproduce a small consistent advantage: at the time, strong-looking corroboration; in hindsight, shared evaluation-block luck.
- Winner: learning rate 10^{-3} , pooled -0.063 bps, $p = 0.0043$ on five seeds.
- No variant reaches the materiality floor (-0.05) with per-seed significance.

Tuning screen, calm regime



- The calm companion: no variant shows a consistent advantage.
- Confirms the regime asymmetry of the development signal.

Tuning and selection summary, all 28 groups (table)

variant	change	regime	valid/total seeds	pooled vs adaptive (bps)	across-seed <i>p</i>
_v3a	learning rate 1e-3	volatile	5/5	-0.0628	0.004
_v3b	learning rate 1e-4	volatile	5/5	-0.0602	0.010
_v2	reward x100	volatile	5/5	-0.0462	0.002
_v1b	network 128x128	volatile	5/5	-0.0562	0.005
_v5	n_steps 8192	volatile	5/5	-0.0485	0.006
_v4a	ent_coef 0.0	volatile	3/3	-0.0472	0.062
_v4b	ent_coef 0.05	volatile	5/5	-0.0267	0.048
_v1a	network 64x64	volatile	2/3	-0.0196	0.151
_v6	reward x100 @ 10M	volatile	2/3	-0.0515	0.150
_v6b	lr 1e-4 @ 10M	volatile	3/3	-0.0395	0.069
_w3a	net128 + reward x100	volatile	3/3	-0.0620	0.016
_w3b	lr 1e-4 + reward x100	volatile	2/3	-0.0363	0.085
_d1	DQN eps 0.05+anneal	volatile	1/3	-0.0403	–
_d2	DQN reward x100+net64	volatile	3/3	-0.0259	0.199
_v3a	learning rate 1e-3	calm	3/3	+0.0013	0.528
_v3b	learning rate 1e-4	calm	2/3	-0.0239	0.111
_v2	reward x100	calm	3/3	-0.0024	0.322
_v1b	network 128x128	calm	3/3	-0.0211	0.140
_v5	n_steps 8192	calm	3/3	-0.0159	0.141
_v4a	ent_coef 0.0	calm	5/5	-0.0213	0.010
_v4b	ent_coef 0.05	calm	3/3	-0.0058	0.200
_v1a	network 64x64	calm	3/3	-0.0159	0.040
_v6	reward x100 @ 10M	calm	3/3	+0.0046	0.705
_v6b	lr 1e-4 @ 10M	calm	3/3	-0.0108	0.214
_w3a	net128 + reward x100	calm	3/3	-0.0063	0.200
_w3b	lr 1e-4 + reward x100	calm	3/3	-0.0116	0.083
_d1	DQN eps 0.05+anneal	calm	1/3	-0.0068	–
_d2	DQN reward x100+net64	calm	0/3	–	–

- 98 runs in total; every group’s pooled result and across-seed consistency.
- The ten-million-step arms reject the “it just needed longer training” explanation.
- The two DQN remediation variants (d1, d2) do not reliably cure the collapse.

Agent configurations (table)

code	agent / variant	change from base
base (PPO)	PPO, discrete actions	net 30×5 , lr $3e-4$, reward $\times 1$, n_steps 2048, ent 0.01, 2M steps
base (DQN)	DQN, discrete actions	net 30×5 , lr $1e-4$, buffer 1M, ϵ $1.0 \rightarrow 0.01$ over 25%
v1a	bigger network	net_arch [64, 64]
v1b	bigger network	net_arch [128, 128]
v2	stronger feedback	reward scale $\times 100$
v3a	faster learning	learning rate $1e-3$
v3b	slower learning	learning rate $1e-4$
v4a	no exploration bonus	ent_coef 0.0
v4b	more exploration	ent_coef 0.05
v5	longer rollout	n_steps 8192
v6	v2 trained longer	reward $\times 100$, 10M steps
v6b	v3b trained longer	lr $1e-4$, 10M steps
w3a	combination	net [128,128] + reward $\times 100$
w3b	combination	lr $1e-4$ + reward $\times 100$
d1	DQN collapse fix	exploration_final_eps 0.05, anneal 0.5
d2	DQN collapse fix	reward $\times 100$ + net [64,64]

- Base PPO and DQN share the same 30×5 network, discount factor and observation set, and matched gradient sample throughput (~ 20 M sample-passes per run): the algorithm comparison is compute-fair.
- Each variant changes exactly one named thing (two for the combination arms).

Full tuning campaign, all 98 runs (appendix table)

- The complete run-level record behind the group summary: every seed, both benchmarks, audit outcome. The table flows over the following pages.
- Included so no result is summarised away; the source files are listed in the provenance table at the end.

algo	regime	variant	seed	vs fixed	<i>p</i>	vs adaptive	valid
dqn	calm	.d1	0	+0.0894	0.002	+0.0895	NO
dqn	calm	.d1	1	-0.0069	0.780	-0.0068	yes
dqn	calm	.d1	2	-0.0068	0.838	-0.0067	NO
dqn	calm	.d2	0	+0.1442	0.002	+0.1443	NO
dqn	calm	.d2	1	+0.1393	0.001	+0.1393	NO
dqn	calm	.d2	2	+0.0099	0.604	+0.0100	NO
dqn	volatile	.d1	0	+0.1398	0.157	+0.1392	NO
dqn	volatile	.d1	1	+0.0324	0.307	+0.0318	NO
dqn	volatile	.d1	2	-0.0396	0.334	-0.0403	yes
dqn	volatile	.d2	0	+0.0063	0.675	+0.0057	yes
dqn	volatile	.d2	1	-0.0729	0.032	-0.0736	yes
dqn	volatile	.d2	2	-0.0092	0.709	-0.0099	yes
ppo	calm	.v1a	0	-0.0226	0.248	-0.0225	yes
ppo	calm	.v1a	1	-0.0068	0.770	-0.0067	yes
ppo	calm	.v1a	2	-0.0185	0.243	-0.0184	yes
ppo	calm	.v1b	0	+0.0053	0.960	+0.0054	yes
ppo	calm	.v1b	1	-0.0246	0.196	-0.0245	yes
ppo	calm	.v1b	2	-0.0441	0.016	-0.0441	yes
ppo	calm	.v2	0	-0.0111	0.732	-0.0110	yes
ppo	calm	.v2	1	+0.0001	0.787	+0.0002	yes
ppo	calm	.v2	2	+0.0036	0.958	+0.0036	yes
ppo	calm	.v3a	0	+0.0337	0.070	+0.0338	yes
ppo	calm	.v3a	1	-0.0230	0.230	-0.0229	yes
ppo	calm	.v3a	2	-0.0069	0.607	-0.0068	yes
ppo	calm	.v3b	0	-0.0327	0.089	-0.0326	yes
ppo	calm	.v3b	1	-0.0153	0.623	-0.0153	yes
ppo	calm	.v3b	2	-0.0148	0.426	-0.0147	NO
ppo	calm	.v4a	0	-0.0189	0.416	-0.0188	yes
ppo	calm	.v4a	1	-0.0126	0.515	-0.0125	yes
ppo	calm	.v4a	2	-0.0298	0.370	-0.0298	yes
ppo	calm	.v4a	3	-0.0382	0.028	-0.0381	yes
ppo	calm	.v4a	4	-0.0076	0.872	-0.0075	yes
ppo	calm	.v4b	0	+0.0027	0.986	+0.0028	yes
ppo	calm	.v4b	1	-0.0161	0.444	-0.0161	yes
ppo	calm	.v4b	2	-0.0043	0.808	-0.0042	yes
ppo	calm	.v5	0	-0.0068	0.808	-0.0067	yes
ppo	calm	.v5	1	-0.0035	0.915	-0.0034	yes
ppo	calm	.v5	2	-0.0377	0.047	-0.0376	yes
ppo	calm	.v6	0	+0.0135	0.671	+0.0136	yes
ppo	calm	.v6	1	+0.0100	0.996	+0.0100	yes
ppo	calm	.v6	2	-0.0099	0.872	-0.0098	yes
ppo	calm	.v6b	0	-0.0319	0.066	-0.0318	yes
ppo	calm	.v6b	1	+0.0047	0.687	+0.0047	yes
ppo	calm	.v6b	2	-0.0053	0.573	-0.0052	yes
ppo	calm	.w3a	0	+0.0017	0.901	+0.0018	yes
ppo	calm	.w3a	1	-0.0178	0.287	-0.0177	yes
ppo	calm	.w3a	2	-0.0029	0.841	-0.0029	yes
ppo	calm	.w3b	0	-0.0218	0.235	-0.0217	yes
ppo	calm	.w3b	1	-0.0102	0.785	-0.0101	yes
ppo	calm	.w3b	2	-0.0032	0.648	-0.0031	yes
ppo	volatile	.v1a	0	-0.0089	0.803	-0.0096	yes
ppo	volatile	.v1a	1	-0.0290	0.374	-0.0296	yes
ppo	volatile	.v1a	2	+0.0232	0.546	+0.0226	NO
ppo	volatile	.v1b	0	-0.0150	0.573	-0.0156	yes
ppo	volatile	.v1b	1	-0.0436	0.196	-0.0443	yes
ppo	volatile	.v1b	2	-0.0640	0.071	-0.0646	yes
ppo	volatile	.v1b	3	-0.0870	0.020	-0.0876	yes
ppo	volatile	.v1b	4	-0.0684	0.051	-0.0691	yes

algo	regime	variant	seed	vs fixed	p	vs adaptive	valid
ppo	volatile	_v2	0	-0.0501	0.168	-0.0507	yes
ppo	volatile	_v2	1	-0.0615	0.119	-0.0621	yes
ppo	volatile	_v2	2	-0.0519	0.186	-0.0525	yes
ppo	volatile	_v2	3	-0.0495	0.156	-0.0502	yes
ppo	volatile	_v2	4	-0.0150	0.965	-0.0156	yes
ppo	volatile	_v3a	0	-0.0494	0.243	-0.0500	yes
ppo	volatile	_v3a	1	-0.1044	0.012	-0.1050	yes
ppo	volatile	_v3a	2	-0.0345	0.369	-0.0352	yes
ppo	volatile	_v3a	3	-0.0798	0.023	-0.0805	yes
ppo	volatile	_v3a	4	-0.0427	0.236	-0.0433	yes
ppo	volatile	_v3b	0	-0.0717	0.047	-0.0724	yes
ppo	volatile	_v3b	1	-0.1019	0.006	-0.1026	yes
ppo	volatile	_v3b	2	-0.0759	0.038	-0.0765	yes
ppo	volatile	_v3b	3	-0.0089	0.866	-0.0095	yes
ppo	volatile	_v3b	4	-0.0394	0.344	-0.0401	yes
ppo	volatile	_v4a	0	-0.0501	0.188	-0.0508	yes
ppo	volatile	_v4a	1	-0.0764	0.018	-0.0771	yes
ppo	volatile	_v4a	2	-0.0131	0.944	-0.0137	yes
ppo	volatile	_v4b	0	-0.0402	0.266	-0.0408	yes
ppo	volatile	_v4b	1	-0.0380	0.334	-0.0387	yes
ppo	volatile	_v4b	2	-0.0231	0.422	-0.0238	yes
ppo	volatile	_v4b	3	-0.0492	0.154	-0.0498	yes
ppo	volatile	_v4b	4	+0.0201	0.682	+0.0195	yes
ppo	volatile	_v5	0	-0.0506	0.156	-0.0512	yes
ppo	volatile	_v5	1	-0.0211	0.698	-0.0217	yes
ppo	volatile	_v5	2	-0.0261	0.454	-0.0267	yes
ppo	volatile	_v5	3	-0.0779	0.056	-0.0785	yes
ppo	volatile	_v5	4	-0.0637	0.124	-0.0643	yes
ppo	volatile	_v6	0	-0.0245	0.829	-0.0252	yes
ppo	volatile	_v6	1	-0.0772	0.083	-0.0778	yes
ppo	volatile	_v6	2	-0.0295	0.608	-0.0302	NO
ppo	volatile	_v6b	0	-0.0620	0.093	-0.0627	yes
ppo	volatile	_v6b	1	-0.0073	0.976	-0.0079	yes
ppo	volatile	_v6b	2	-0.0474	0.151	-0.0480	yes
ppo	volatile	_w3a	0	-0.0602	0.100	-0.0609	yes
ppo	volatile	_w3a	1	-0.0814	0.029	-0.0821	yes
ppo	volatile	_w3a	2	-0.0423	0.470	-0.0430	yes
ppo	volatile	_w3b	0	-0.0456	0.153	-0.0462	yes
ppo	volatile	_w3b	1	+0.0471	0.467	+0.0465	NO
ppo	volatile	_w3b	2	-0.0257	0.647	-0.0263	yes

Training curves



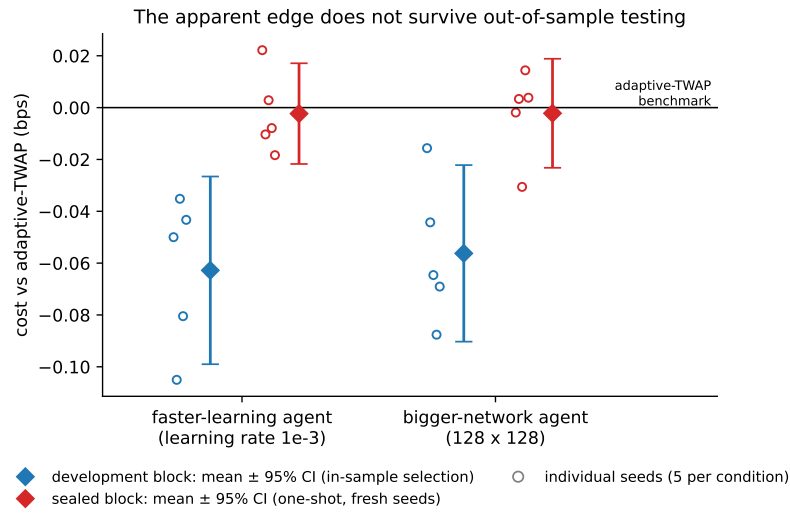
- Base and selected configurations plateau; training ran to completion at 2M steps.
- Measured on the monitor block, which is never used for decisions.
- No sign that the null is under-training in disguise.

Both sealed confirmations (table)

config	sealed block	seed	vs fixed	vs adaptive	p (adaptive)	valid
faster-learning (v3a)	9,000,000	5	+0.0221	+0.0222	0.749	yes
		6	-0.0104	-0.0103	0.813	yes
		7	+0.0028	+0.0028	0.804	yes
		8	-0.0080	-0.0079	0.936	yes
		9	-0.0184	-0.0184	0.758	yes
	pooled vs adaptive = -0.0023 bps; across-seed $p = 0.379$; cheaper in 3/5; PASS = FALSE					
bigger-network (v1b)	13,000,000	10	-0.0010	-0.0019	0.650	yes
		11	+0.0042	+0.0033	0.953	yes
		12	-0.0297	-0.0306	0.294	yes
		13	+0.0153	+0.0144	0.491	yes
		14	+0.0047	+0.0038	0.961	yes
	pooled vs adaptive = -0.0022 bps; across-seed $p = 0.394$; cheaper in 2/5; PASS = FALSE					

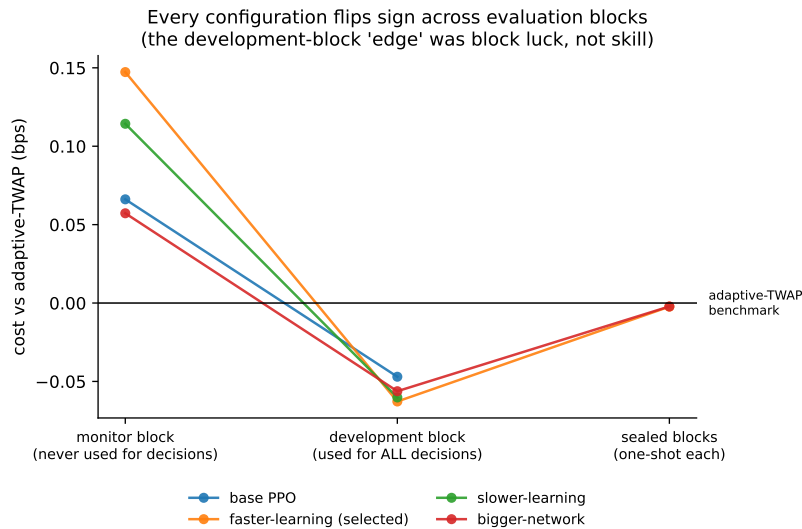
- Two champions (one chosen on volatile-regime performance, one chosen by the pre-registered selection rule applied exactly as written), five fresh seeds each, evaluated once on separate never-used sealed blocks.
- Results: -0.0023 bps ($p = 0.379$, 3/5 cheaper) and -0.0022 bps ($p = 0.394$, 2/5 cheaper); both confidence intervals bracket zero.
- Testing the second candidate also resolved a disclosed selection-metric ambiguity empirically: both picks are approximately zero out of sample.
- The confirmation family was capped at two tests in advance; no re-runs.

The headline figure



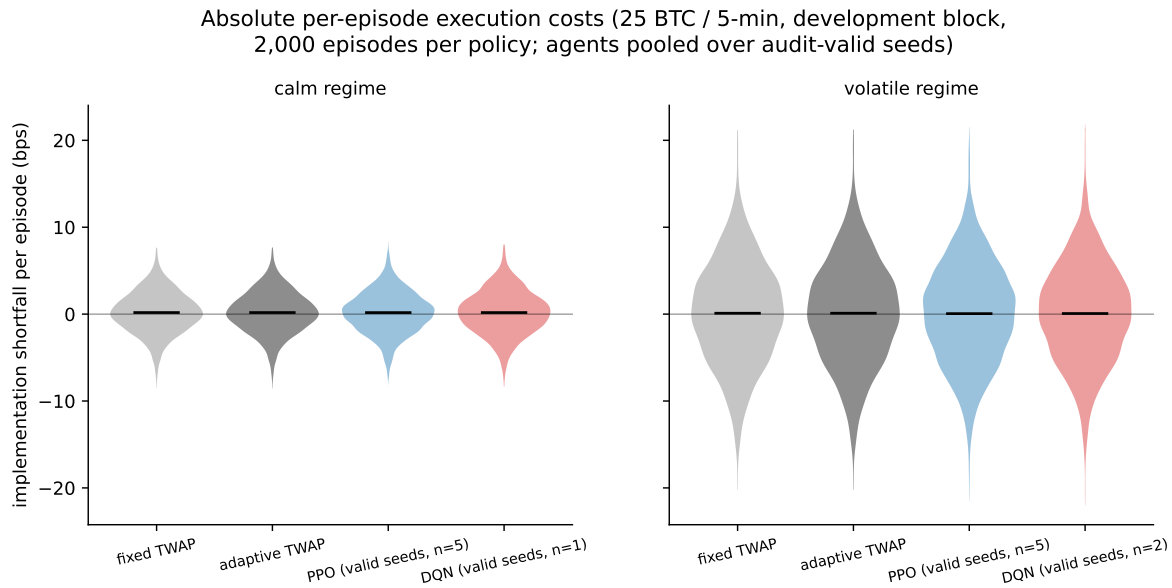
- Development-block advantage versus sealed reality, both champions side by side.
- The signal that was consistent across five seeds and twelve variants does not exist on fresh data.
- Headline of the dissertation: a boundary null, established twice under pre-committed rules.

Why the illusion is visible in hindsight



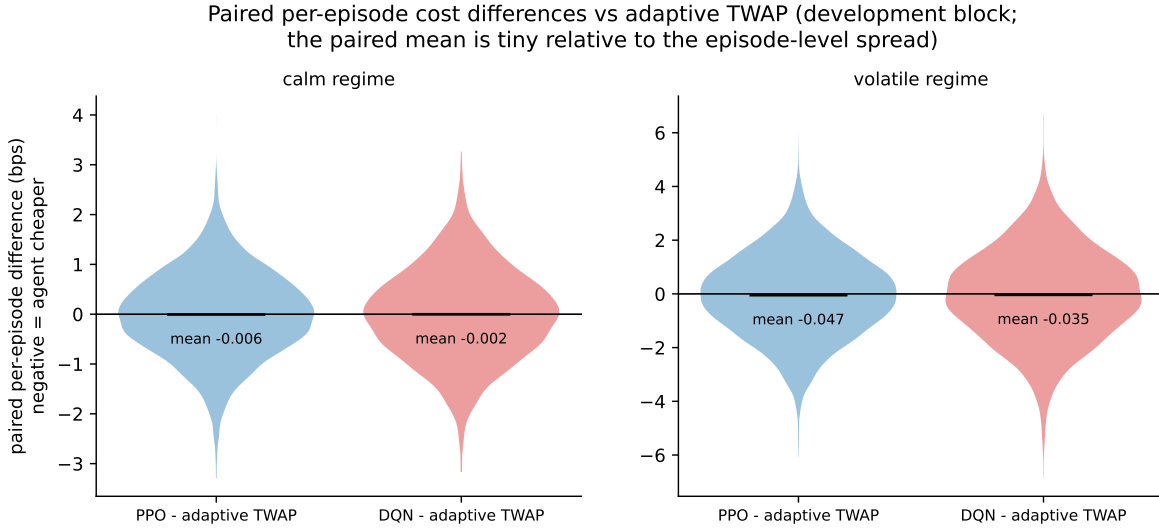
- The same configurations on three disjoint blocks: worse than TWAP on the monitor block, better on the development block, zero on sealed blocks.
- Seed agreement cannot detect shared evaluation-set luck: every seed is graded on the same episodes.
- This is demonstration number two of the spurious-edge anatomy (the drift was number one; the grid triggers below are number three).

Absolute cost distributions



- What each shape is: one violin summarises the 2,000 individual episode costs of one policy; the wider the shape at a given height, the more episodes cost that amount.
- How to read the axis: the vertical axis is the cost of executing the 25 BTC order in basis points relative to the price at the start of the episode; positive means the policy paid more than the arrival price, negative means it paid less.
- The black bar inside each violin is that policy's average cost; all four averages sit essentially at zero in both panels.
- The key comparison: within each panel the four shapes (fixed TWAP, adaptive TWAP, PPO, DQN) are nearly identical, so the choice of policy barely changes what execution costs; the market decides, not the policy.
- Regime effect: the volatile panel's shapes span roughly -20 to $+20$ basis points versus roughly -8 to $+8$ in calm, so volatility widens the cost distribution by about a factor of 2.5 for every policy equally (sd 2.35 to 5.88 bps).
- Trust in the data: this replay provably describes the same evaluations as the sealed verdicts, because every recomputed average matches the sealed record exactly (20 of 20 runs).

Paired per-episode differences



- The paired agent-minus-benchmark differences, pooled over audit-valid seeds.
- The paired mean (-0.047 at best) is three orders of magnitude smaller than the episode-level spread.
- This is the visual justification for 2,000 paired episodes and strict statistical discipline: anything less and noise decides.

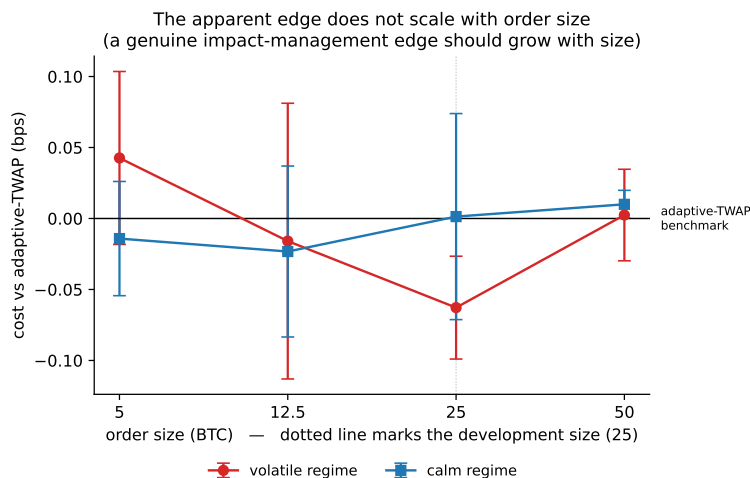
Descriptive statistics (table)

regime	policy	mean	sd	median	5%	95%	n episodes
calm	fixed TWAP	+0.1803	2.3546	+0.1840	-3.7238	+4.0427	2,000
calm	adaptive TWAP	+0.1802	2.3540	+0.1810	-3.7212	+4.0245	2,000
calm	PPO (valid seeds)	+0.1747	2.3661	+0.2089	-3.7477	+3.9982	10,000
calm	DQN (valid seeds)	+0.1787	2.4817	+0.1849	-3.9363	+4.2143	2,000
volatile	fixed TWAP	+0.1031	5.8800	+0.1013	-9.7930	+9.4489	2,000
volatile	adaptive TWAP	+0.1037	5.8793	+0.0922	-9.7133	+9.4541	2,000
volatile	PPO (valid seeds)	+0.0567	5.7882	+0.1640	-9.3214	+9.2684	10,000
volatile	DQN (valid seeds)	+0.0688	5.6446	+0.1161	-9.0810	+9.1627	4,000

- Mean, spread, median and 5/95 percentiles per policy and regime.
- Medians sit near zero everywhere; tails are wide and symmetric; no policy shifts the distribution visibly.

3. Robustness: the null holds everywhere

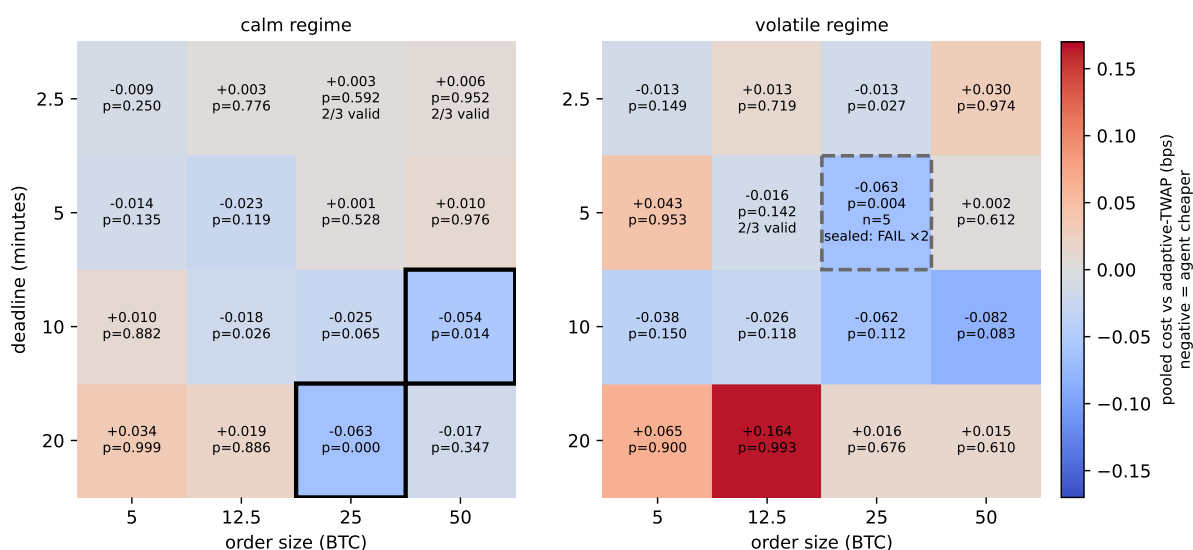
Size response



- A genuine impact-management edge should grow with order size.
- Instead the signal exists only at the 25 BTC development size and is absent at 5, 12.5 and 50 BTC: non-monotone, the signature of selection on noise.
- This was the fourth independent line of evidence against the development edge, before the grid.

The full grid

Robustness grid, development-block evidence: two calm cells (solid outline) meet the pre-registered follow-up trigger; the centre volatile signal (dashed) already failed two sealed tests



- 66 new runs completing sizes {5, 12.5, 25, 50} BTC by deadlines {2.5, 5, 10, 20} minutes, both regimes; integrity-checked before judging.
- Null in 20 of 22 new groups; several groups significantly worse than TWAP (e.g. 12.5 BTC at 20 minutes, volatile: +0.164 bps).

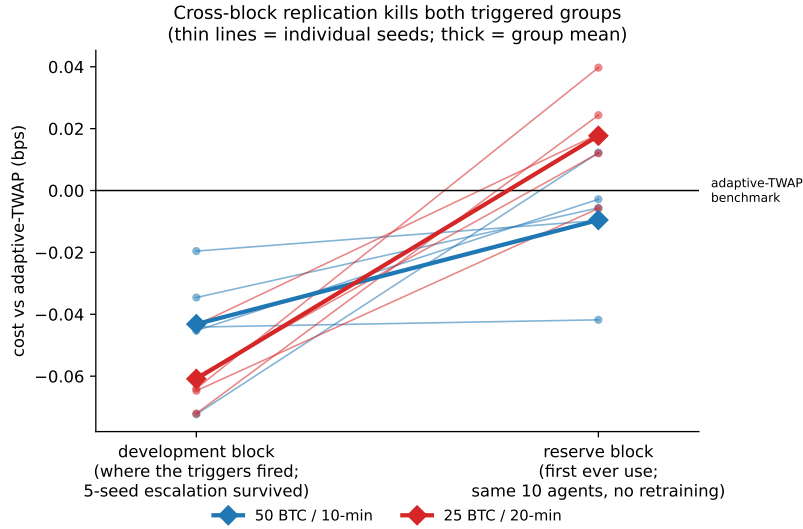
- Two calm cells (solid outline) met the pre-registered follow-up trigger; the centre volatile cell (dashed) is the already-falsified original signal.

Grid summary (table)

regime	deadline (min)	size (BTC)	valid/ total	cheaper	pooled vs adaptive (bps)	across-seed p	follow-up trigger
calm	2.5	5	3/3	2/3	-0.0094	0.2498	no
	2.5	12.5	3/3	1/3	+0.0032	0.7761	no
	2.5	25	2/3	1/2	+0.0029	0.5919	no
	2.5	50	2/3	0/2	+0.0060	0.9525	no
	5	5	3/3	2/3	-0.0142	0.1346	no
	5	12.5	3/3	2/3	-0.0233	0.1191	no
	5	25	3/3	2/3	+0.0013	0.5279	no
	5	50	3/3	0/3	+0.0100	0.9757	no
	10	5	3/3	1/3	+0.0105	0.8819	no
	10	12.5	3/3	3/3	-0.0178	0.0261	no
	10	25	3/3	3/3	-0.0250	0.0646	no
	10	50	3/3	3/3	-0.0539	0.0140	YES
	20	5	3/3	0/3	+0.0338	0.9991	no
	20	12.5	3/3	1/3	+0.0189	0.8863	no
	20	25	3/3	3/3	-0.0629	0.0003	YES
	20	50	3/3	2/3	-0.0169	0.3466	no
volatile	2.5	5	3/3	2/3	-0.0132	0.1492	no
	2.5	12.5	3/3	1/3	+0.0126	0.7186	no
	2.5	25	3/3	3/3	-0.0134	0.0273	no
	2.5	50	3/3	0/3	+0.0301	0.9740	no
	5	5	3/3	0/3	+0.0426	0.9526	no
	5	12.5	2/3	2/2	-0.0159	0.1424	no
	5	25	5/5	5/5	-0.0628	0.0043	closed (failed both sealed tests)
	5	50	3/3	1/3	+0.0024	0.6120	
	10	5	3/3	2/3	-0.0383	0.1503	no
	10	12.5	3/3	3/3	-0.0262	0.1180	no
	10	25	3/3	2/3	-0.0619	0.1123	no
	10	50	3/3	3/3	-0.0821	0.0833	no
	20	5	3/3	0/3	+0.0648	0.8996	no
	20	12.5	3/3	0/3	+0.1638	0.9926	no
	20	25	3/3	2/3	+0.0156	0.6764	no
	20	50	3/3	1/3	+0.0150	0.6099	no

- All 16 cells per regime with pooled results, consistency, and trigger status.
- Roughly one false trigger was expected among 22 groups by chance; two were observed, hence follow-up rather than claims.

The replication ladder kills both triggers



- The triggers: $p = 0.014$, and $p = 0.0003$ with three seeds within 0.005 bps of one another.
- Both survived escalation to five seeds on the development block ($p = 0.0037$ and $p = 0.0001$).
- Both then died on the first-ever use of the reserve block: -0.0095 ($p = 0.17$) and $+0.0177$ ($p = 0.96$), the second reversing sign entirely.
- Cost of catching it: four training runs. Sealed tests spent: zero.
- Third demonstrated evaluation-set illusion, and the cleanest.

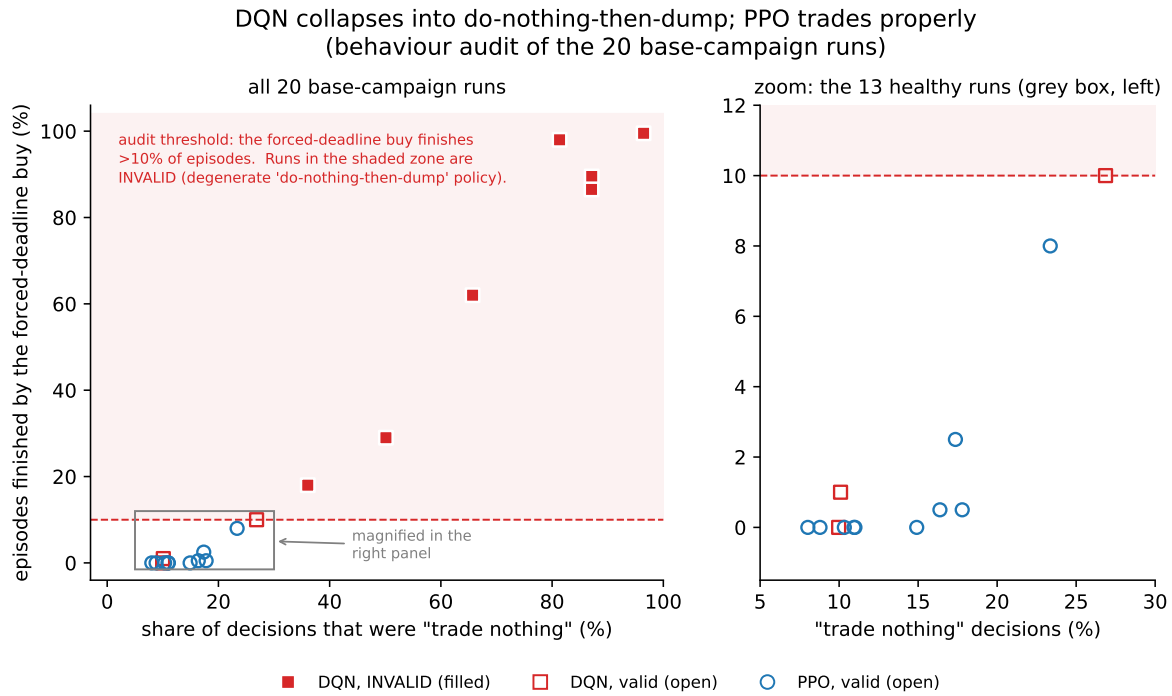
Ladder detail (table)

group	seed	development (bps)	reserve (bps)	change
50 BTC / 10-min	0	-0.0453	-0.0028	+0.0425
	1	-0.0723	+0.0122	+0.0846
	2	-0.0441	-0.0418	+0.0023
	3	-0.0346	-0.0056	+0.0290
	4	-0.0196	-0.0097	+0.0099
	pooled	-0.0432	-0.0095	+0.0336
25 BTC / 20-min	0	-0.0601	+0.0120	+0.0721
	1	-0.0640	+0.0397	+0.1037
	2	-0.0647	-0.0058	+0.0589
	3	-0.0721	+0.0244	+0.0964
	4	-0.0435	+0.0182	+0.0617
	pooled	-0.0609	+0.0177	+0.0786

- Every escalated seed's development and reserve result side by side.
- Note the tight development agreement dissolving into scatter around zero on fresh episodes.

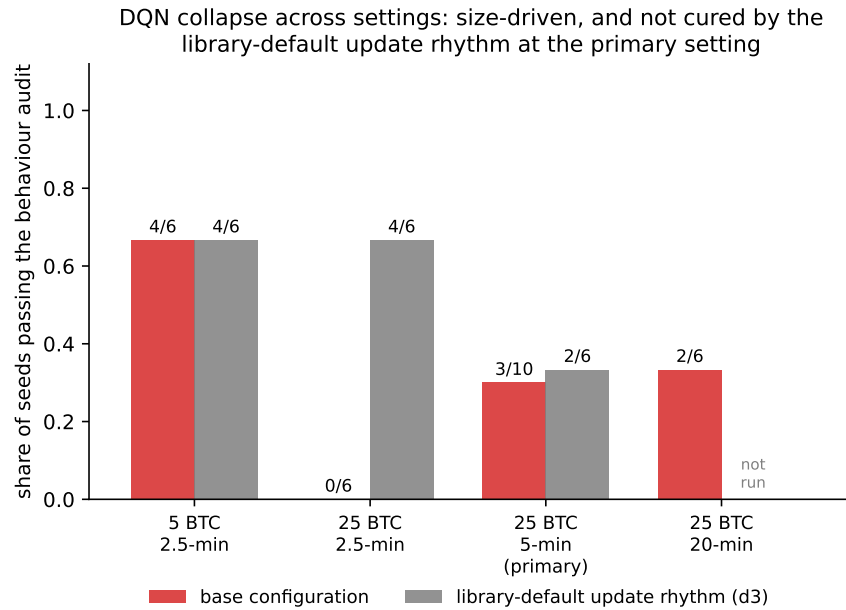
4. Why DQN fails

The collapse at the primary setting



- The behaviour audit runs before any cost comparison: an agent whose forced deadline purchase completes more than 10% of episodes, or that is more than 90% inactive, is invalid.
- Seven of ten DQN agents collapse into inaction followed by the forced purchase; all ten PPO agents trade properly.
- Collapsed agents' cost numbers are excluded from verdicts but always reported.

Systematic, size-driven, and not a configuration artifact



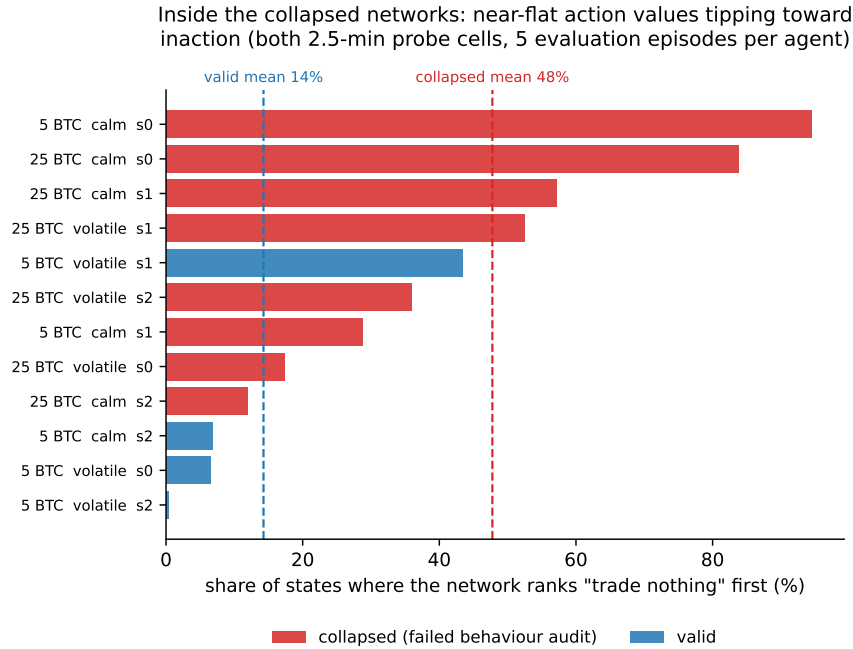
- Pre-registered probe across settings: mostly healthy at 5 BTC (4/6 valid), total collapse at 25 BTC even at the shortest deadline (0/6), majority collapse at every 25 BTC deadline.
- So the driver is order size (reward dominated by the agent's own impact), not idle time; the going-in idle-room hypothesis was refuted by the probe's own design.
- The collapse concentrates in the calm regime (1 of 9 calm probe agents valid versus 5 of 9 volatile).
- Grey bars: retraining with the library-default update rhythm (18 runs, pre-registered) leaves the primary setting unchanged; the rhythm objection is closed with data.

DQN probe detail (table)

cell	regime	seed	do-nothing share	forced-buy share	vs adaptive (bps)	<i>p</i>	audit
5 BTC / 2.5-min	calm	0	93%	43%	-0.0287	0.425	COLL.
5 BTC / 2.5-min	calm	1	36%	22%	-0.0250	0.159	COLL.
5 BTC / 2.5-min	calm	2	6%	2%	+0.0036	0.824	valid
5 BTC / 2.5-min	volatile	0	7%	0%	+0.0316	0.309	valid
5 BTC / 2.5-min	volatile	1	43%	0%	-0.0540	0.176	valid
5 BTC / 2.5-min	volatile	2	1%	0%	-0.0406	0.041	valid
25 BTC / 2.5-min	calm	0	82%	100%	+0.0258	0.500	COLL.
25 BTC / 2.5-min	calm	1	57%	18%	-0.0089	0.646	COLL.
25 BTC / 2.5-min	calm	2	13%	67%	-0.0223	0.179	COLL.
25 BTC / 2.5-min	volatile	0	17%	48%	-0.0410	0.529	COLL.
25 BTC / 2.5-min	volatile	1	46%	12%	-0.0523	0.247	COLL.
25 BTC / 2.5-min	volatile	2	37%	72%	-0.0106	0.828	COLL.
25 BTC / 20-min	calm	0	43%	54%	-0.0518	0.088	COLL.
25 BTC / 20-min	calm	1	43%	87%	-0.0019	0.866	COLL.
25 BTC / 20-min	calm	2	54%	70%	-0.0462	0.346	COLL.
25 BTC / 20-min	volatile	0	16%	8%	-0.2288	0.025	valid
25 BTC / 20-min	volatile	1	30%	0%	-0.0539	0.427	valid
25 BTC / 20-min	volatile	2	25%	78%	-0.0193	0.570	COLL.

- All 18 probe runs: inaction share, forced-purchase share, costs, audit outcome.
- Even where DQN survives the audit, no cost advantage exists anywhere.

The mechanism, read from the networks

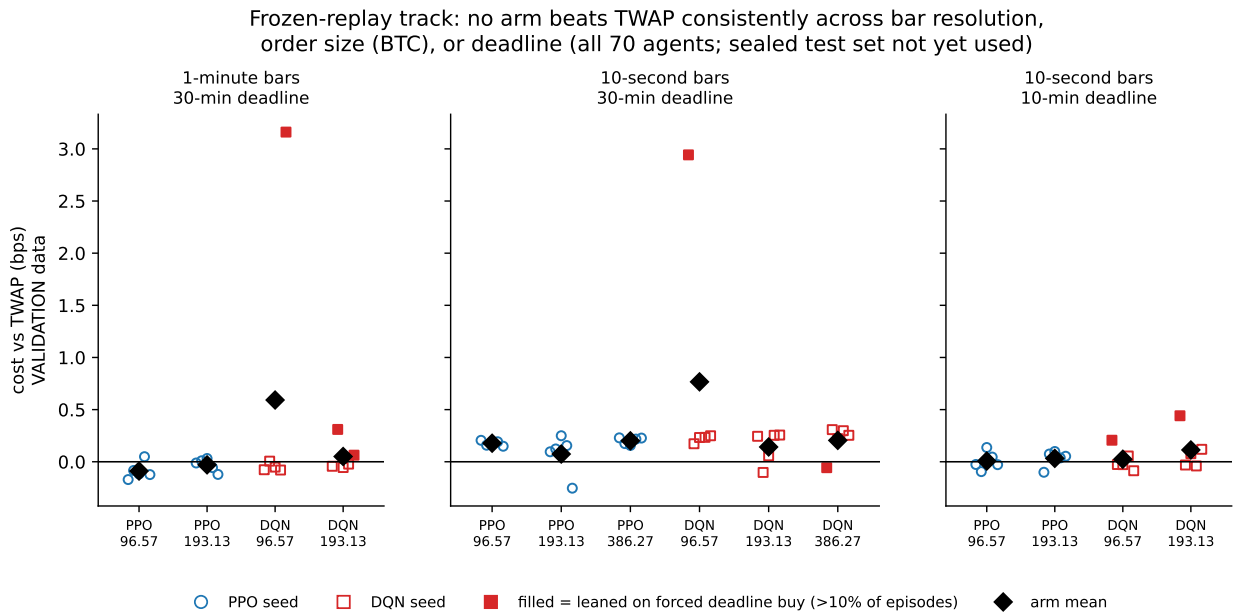


- Direct inspection of the trained networks over evaluation states.

- Across all agents, learned action-value separations (0.006 to 0.012) sit an order of magnitude below per-step reward noise (sd 0.025 to 0.057): the value signal is at the edge of resolvability for everyone.
- What distinguishes collapse is where the near-flat surface tips: collapsed agents rank “trade nothing” first in 48% of states versus 14% for healthy agents.
- Interpretation (labelled as such): PPO’s batch-normalised advantages amplify microscopic differences; DQN regresses raw-scale values.
- Scoped claim: this configuration on impact-dominated rewards; Double-DQN and prioritised replay are named future work.

5. The frozen-replay track (validation results; sealed exam pending)

Three-lever null



- All 70 agents across bar resolution (1-minute, 10-second), order size (0.5%, 1%, 2% of daily volume) and deadline (30, 10 minutes).
- No arm beats TWAP consistently; on 10-second bars at 30 minutes the agents are consistently worse.
- The best-looking arm (PPO, 96.57 BTC, 1-minute bars: -0.089 , 4/5 cheaper) is within chance expectation for 14 screened arms and receives no special treatment: every arm sits the same one-shot sealed exam.
- Six DQN runs show the same deadline-leaning failure as the reactive track (filled markers).
- Integrity notes: four legacy stub runs were exposed by a census, retracted and retrained (the arm’s true figure is three times weaker than the stub-based one); fill-in comparability was proven by a bit-exact sibling retrain.

Frozen-replay arm summary (table)

panel	arm	pooled vs TWAP (bps)	cheaper seeds	flagged seeds	across-seed <i>p</i>
1-min bars, 30-min	PPO 96.57	-0.0890	4/5	0/5	0.037
1-min bars, 30-min	PPO 193.13	-0.0297	3/5	0/5	0.167
1-min bars, 30-min	DQN 96.57	+0.5920	3/5	1/5	0.796
1-min bars, 30-min	DQN 193.13	+0.0505	3/5	2/5	0.750
10-s bars, 30-min	PPO 96.57	+0.1779	0/5	0/5	1.000
10-s bars, 30-min	PPO 193.13	+0.0740	1/5	0/5	0.781
10-s bars, 30-min	PPO 386.27	+0.2010	0/5	0/5	1.000
10-s bars, 30-min	DQN 96.57	+0.7663	0/5	1/5	0.884
10-s bars, 30-min	DQN 193.13	+0.1417	1/5	0/5	0.940
10-s bars, 30-min	DQN 386.27	+0.2049	1/5	1/5	0.981
10-s bars, 10-min	PPO 96.57	+0.0066	3/5	0/5	0.562
10-s bars, 10-min	PPO 193.13	+0.0324	1/5	0/5	0.797
10-s bars, 10-min	DQN 96.57	+0.0244	3/5	1/5	0.672
10-s bars, 10-min	DQN 193.13	+0.1133	2/5	1/5	0.867

- Pooled result, seeds cheaper, flagged seeds and consistency for all 14 arms.
- Validation data by design; test columns are added by the sealed exam.

The size lever

The size lever: execution gets harder as orders grow, but no exploitable gap opens



- Benchmark execution cost grows steeply with order size (to ~13bps at 5% of daily volume): the execution problem gets genuinely harder.
- No exploitable agent advantage opens at any trained size; one collapsed DQN seed is marked off scale.

Frozen-replay per-run detail (appendix table)

- Every one of the 70 agents: validation result and deadline-leaning share. The table flows over the following pages.
- DL marks the forced deadline purchase completing more than 10% of episodes.

panel	arm	seed	vs TWAP (bps)	resid
1-min bars, 30-min	PPO 96.57	0	-0.1707	0.8%
1-min bars, 30-min	PPO 96.57	1	-0.0834	0.2%
1-min bars, 30-min	PPO 96.57	2	-0.1169	0.5%
1-min bars, 30-min	PPO 96.57	3	+0.0487	1.0%
1-min bars, 30-min	PPO 96.57	4	-0.1230	0.5%
1-min bars, 30-min	PPO 193.13	0	-0.0118	2.0%
1-min bars, 30-min	PPO 193.13	1	+0.0094	0.2%
1-min bars, 30-min	PPO 193.13	2	+0.0314	1.8%
1-min bars, 30-min	PPO 193.13	3	-0.0562	1.2%
1-min bars, 30-min	PPO 193.13	4	-0.1214	0.8%
1-min bars, 30-min	DQN 96.57	0	-0.0764	0.0%
1-min bars, 30-min	DQN 96.57	1	+0.0060	0.5%
1-min bars, 30-min	DQN 96.57	2	-0.0512	0.8%
1-min bars, 30-min	DQN 96.57	3	-0.0793	0.0%
1-min bars, 30-min	DQN 96.57	4	+3.1610	97.8% DL
1-min bars, 30-min	DQN 193.13	0	-0.0433	1.5%
1-min bars, 30-min	DQN 193.13	1	+0.3100	36.8% DL
1-min bars, 30-min	DQN 193.13	2	-0.0553	0.2%
1-min bars, 30-min	DQN 193.13	3	-0.0224	1.5%
1-min bars, 30-min	DQN 193.13	4	+0.0632	11.8% DL
10-s bars, 30-min	PPO 96.57	0	+0.2056	0.0%
10-s bars, 30-min	PPO 96.57	1	+0.1587	0.0%
10-s bars, 30-min	PPO 96.57	2	+0.1846	0.2%
10-s bars, 30-min	PPO 96.57	3	+0.1919	0.0%
10-s bars, 30-min	PPO 96.57	4	+0.1488	0.2%
10-s bars, 30-min	PPO 193.13	0	+0.0956	0.2%
10-s bars, 30-min	PPO 193.13	1	+0.1229	0.2%
10-s bars, 30-min	PPO 193.13	2	+0.2496	0.0%
10-s bars, 30-min	PPO 193.13	3	+0.1552	0.5%
10-s bars, 30-min	PPO 193.13	4	-0.2533	5.5%
10-s bars, 30-min	PPO 386.27	0	+0.2294	0.2%
10-s bars, 30-min	PPO 386.27	1	+0.1743	0.0%
10-s bars, 30-min	PPO 386.27	2	+0.1565	0.0%
10-s bars, 30-min	PPO 386.27	3	+0.2174	0.0%
10-s bars, 30-min	PPO 386.27	4	+0.2272	0.0%
10-s bars, 30-min	DQN 96.57	0	+2.9426	97.8% DL
10-s bars, 30-min	DQN 96.57	1	+0.1728	7.8%
10-s bars, 30-min	DQN 96.57	2	+0.2327	0.0%
10-s bars, 30-min	DQN 96.57	3	+0.2341	0.5%
10-s bars, 30-min	DQN 96.57	4	+0.2493	0.0%
10-s bars, 30-min	DQN 193.13	0	+0.2441	0.0%
10-s bars, 30-min	DQN 193.13	1	-0.1025	10.0%
10-s bars, 30-min	DQN 193.13	2	+0.0570	0.5%
10-s bars, 30-min	DQN 193.13	3	+0.2537	0.2%
10-s bars, 30-min	DQN 193.13	4	+0.2561	0.5%
10-s bars, 30-min	DQN 386.27	0	-0.0569	27.0% DL
10-s bars, 30-min	DQN 386.27	1	+0.3082	1.0%
10-s bars, 30-min	DQN 386.27	2	+0.2214	1.0%
10-s bars, 30-min	DQN 386.27	3	+0.2978	0.0%
10-s bars, 30-min	DQN 386.27	4	+0.2537	0.0%
10-s bars, 10-min	PPO 96.57	0	-0.0262	0.2%
10-s bars, 10-min	PPO 96.57	1	-0.0946	0.2%
10-s bars, 10-min	PPO 96.57	2	+0.1367	0.0%
10-s bars, 10-min	PPO 96.57	3	+0.0450	0.0%
10-s bars, 10-min	PPO 96.57	4	-0.0280	0.5%
10-s bars, 10-min	PPO 193.13	0	-0.1008	1.0%
10-s bars, 10-min	PPO 193.13	1	+0.0741	1.0%
10-s bars, 10-min	PPO 193.13	2	+0.0985	1.5%

panel	arm	seed	vs TWAP (bps)	resid
10-s bars, 10-min	PPO 193.13	3	+0.0373	1.5%
10-s bars, 10-min	PPO 193.13	4	+0.0529	0.5%
10-s bars, 10-min	DQN 96.57	0	+0.2065	26.2% DL
10-s bars, 10-min	DQN 96.57	1	-0.0264	0.0%
10-s bars, 10-min	DQN 96.57	2	-0.0275	0.0%
10-s bars, 10-min	DQN 96.57	3	+0.0553	0.2%
10-s bars, 10-min	DQN 96.57	4	-0.0861	1.8%
10-s bars, 10-min	DQN 193.13	0	+0.4411	48.8% DL
10-s bars, 10-min	DQN 193.13	1	-0.0312	3.0%
10-s bars, 10-min	DQN 193.13	2	+0.0772	0.0%
10-s bars, 10-min	DQN 193.13	3	-0.0401	0.0%
10-s bars, 10-min	DQN 193.13	4	+0.1193	0.0%

6. Committed next steps

- Frozen-replay sealed exam: the evaluator must first reproduce known validation numbers exactly; then one shot over all 70 agents, all arms reported, multiplicity stated; gated.
- Measured-signal extension: measure the order-flow to future-return relationship from the L4 data and inject exactly that into the reactive simulator; the one experiment that can change the headline; pre-registered before running.
- Liquidity and volume profiles (feeds a VWAP benchmark); Almgren-Chriss and VWAP added to the benchmark set; attribution (research question 3) after the extension verdict; decision-frequency check (12 runs at 0.5s and 2s).

7. Provenance (source of truth for every number)

result	source of truth
Primary campaign	oxford_14/step5.v3/
Tuning + selection	oxford_14/step5.selection.v3/
Sealed confirmations	oxford_14/step5.confirm_{v3a,v1b}/
Sweeps + grid	oxford_14/step5.sweep_*, step5_grid_*/
Replication ladder	oxford_14/step5.esc_*, step5_xblock_*/
DQN probe + d3	oxford_14/step5.dqnprobe_*, step5_d3_*/
Q-value diagnostic	reports/diagnostics/dqn_q_flatness.json
Per-episode distributions	oxford_14/per_episode.v3/
Environment gates	oxford_14/step4.gates.v3.json, step3g/fairness_*.json
Frozen-replay agents	runs/, runs_10s/, runs_10s_10min/ meta.json
Frozen-replay benchmark sweep	size_sweep_results.json

- Every number in this document is generated by script from these archived files; nothing is transcribed by hand.
- Markdown summaries in the project documentation are convenience copies; these artifacts are authoritative.